Lab 8: Cellular Genetics — Mitosis and Meiosis

BIOL-1

Name: ______________________________ **Date:** ______________________________

Learning Objectives

By the end of this lab, you will be able to:

1. **Define** key vocabulary: diploid, haploid, homologous chromosomes, sister chromatids, allele, gamete.
2. **Explain** what happens to chromosomes during interphase (S phase) before division begins.
3. **Model mitosis** step by step using paper chromosomes, tracing each allele through every phase.
4. **Model meiosis I and II** using paper chromosomes, distinguishing what separates at each stage.
5. **Explain** the critical difference between Anaphase I (homologs separate) and Anaphase II (sister chromatids separate).
6. **Compare and contrast** mitosis and meiosis by purpose, number of divisions, chromosome number, genetic outcome, and other features.
7. **Explain** why meiosis produces genetically unique gametes while mitosis produces identical daughter cells.

Overview

Every cell in your body came from a single fertilized egg that divided over and over again.

That process — **cell division** — is how you grow, repair wounds, and replace worn-out cells. But not all cell division works the same way:

```
                CELL DIVISION
                      |
           ┌──────────┴──────────┐
        MITOSIS               MEIOSIS
   (growth & repair)       (reproduction)
         |                       |
```

```
   1 division            2 divisions
        |                      |
 2 diploid cells        4 haploid cells
  (identical)          (genetically unique)
```

In this lab, you will use **paper chromosomes** — RED for maternal, BLUE for paternal — to physically walk through both processes and see exactly where each allele ends up.

Key Terms

Term	Definition
Diploid (2n)	A cell with two full sets of chromosomes (one from each parent)
Haploid (n)	A cell with one set of chromosomes
Homologous chromosomes	Matching chromosome pairs — one maternal (red), one paternal (blue)
Sister chromatids	Two identical copies of one chromosome, joined at the centromere after DNA replication
Allele	A version of a gene (e.g., **A** = dominant, **a** = recessive)
Centromere	The region where sister chromatids are joined together
Gamete	A sex cell (egg or sperm), produced by meiosis

Materials

- **Chromosome Cutouts** (last page of this lab)
- Chromosome Pair 1: Gene A (alleles **A** and **a**)
- Chromosome Pair 2: Gene B (alleles **B** and **b**)
- Blank paper for staging (draw large circles to represent cells)

> ***No scissors needed.*** *Carefully tear along the dashed borders of each chromosome card. Slow, steady tears work best — go corner to corner.*

Part 1: Chromosome Basics

Before you tear out the chromosomes, answer these questions to check your vocabulary.

1. Match each term to its definition:

Term	Definition
Chromatin	
Chromosome	
Sister chromatids	
Homologous chromosomes	
Diploid (2n)	
Haploid (n)	

2. What is the difference between sister chromatids and homologous chromosomes?

Setting Up Your Cell

We are modeling a simple organism with **2 pairs of chromosomes** (2n = 4). The genotype of your starting cell is **AaBb**.

```
Chromosome Pair 1 (Gene A):
┌──────────┐   ┌──────────┐
│ Maternal │   │ Paternal │
│    A     │   │    a     │
│  [RED]   │   │  [BLUE]  │
└──────────┘   └──────────┘

Chromosome Pair 2 (Gene B):
┌──────────┐   ┌──────────┐
│ Maternal │   │ Paternal │
│    B     │   │    b     │
```

```
│   [RED]      │       │  [BLUE]   │
└──────────────┘       └───────────┘

Starting cell genotype: AaBb   (diploid, 2n = 4)
```

Procedure

1. Tear out the last page and cut out the 4 **Starting Cell** chromosomes.
2. Draw a large circle on blank paper to represent your cell.
3. Place all 4 chromosomes inside the circle. This is your **diploid parent cell (2n = 4)**.

3. Record the genotype of your starting cell:

[ ]

4. How many chromosomes are in this cell? Is it diploid or haploid?

[ ]

Part 2: Mitosis — Making Identical Copies

Mitosis is used for growth, repair, and replacing old cells. It produces two genetically identical diploid cells.

Step 1: S Phase (DNA Replication)

Before mitosis begins, each chromosome is copied.

1. Cut out the 4 **S-Phase Duplicate** chromosomes from the cutouts page.
2. **Tape** each duplicate to its matching original at the center — these become **sister chromatids**.
3. You now have 4 duplicated chromosomes (8 chromatids total) inside one cell.

```
After S Phase — each chromosome is now an X-shaped pair of sister chromatids.
On your desk it should look like this:

┌──────╥──────┐  ┌──────╥──────┐  ┌──────╥──────┐  ┌──────╥──────┐
│  A   ║  A   │  │  a   ║  a   │  │  B   ║  B   │  │  b   ║  b   │
│ RED  ║ RED  │  │BLUE  ║BLUE  │  │ RED  ║ RED  │  │BLUE  ║BLUE  │
└──────╨──────┘  └──────╨──────┘  └──────╨──────┘  └──────╨──────┘
```

Step 2: Prophase

1. Chromosomes condense into visible X-shapes (sister chromatids joined at centromere).
2. Arrange all 4 duplicated chromosomes loosely inside your cell circle.
3. **Note:** In mitosis, homologous chromosomes do **not** pair up.

Step 3: Metaphase

1. **Line up** all 4 duplicated chromosomes single-file along the middle of the cell (the metaphase plate).

Step 4: Anaphase

1. **Separate the sister chromatids** — pull each taped pair apart.
2. Move one chromatid from each pair to one side of the cell, the other to the opposite side.
3. Each side should now have 4 chromatids (one A, one a, one B, one b).

Step 5: Telophase & Cytokinesis

1. **Draw a line** down the middle to separate the two groups into two daughter cells.
2. Each daughter cell now has 4 chromosomes (AaBb, 2n = 4) — identical to the parent.

Mitosis Results — Allele Tracking

#	Cell	Gene A Allele(s)	Gene B Allele(s)	Full Genotype	Diploid or Haploid?	# Chromosomes
1						
2						
3						

5. Are the two daughter cells genetically identical to each other and to the parent cell?

6. Did the chromosome number change? Explain what happened.

7. Why is mitosis described as producing "clones" of the parent cell?

Part 3: Meiosis I — Separating Homologous Chromosomes

Reset your cell: gather all chromosomes back together. Re-tape sister chromatids if needed. You should again have 4 duplicated chromosomes inside one diploid cell (AaBb, 2n = 4).

Step 1: Prophase I

Chromosomes condense. **Homologous chromosomes pair up** side by side — this is called **synapsis**.

1. Bring the A chromosome (red) next to the a chromosome (blue) — this is one **homologous pair**.
2. Bring the B chromosome (red) next to the b chromosome (blue) — this is the other **homologous pair**.
3. Each pair is called a **tetrad** (4 chromatids in close contact).

(Note: In real cells, crossing over can happen here — homologs exchange segments of DNA. We are keeping it simple and skipping that step today.)

Step 2: Metaphase I — Homolog Pairs Line Up

1. **Line up the two tetrads** (homologous pairs) along the middle of the cell as pairs, not as individual chromosomes.

Step 3: Anaphase I

1. **Separate the homologous pairs** — pull each tetrad apart into its two individual (still-duplicated) chromosomes.
2. Move one complete chromosome from each pair to each side.
3. **Important:** Sister chromatids stay together! Only homologs separate here.

Step 4: Telophase I & Cytokinesis I

1. **Divide the cell** into two new cells (draw two new circles).
2. Each new cell has **2 duplicated chromosomes** (sister chromatids still joined).
3. Each cell is now **haploid (n = 2)**, but chromosomes are still duplicated.

Meiosis I Results — Allele Tracking

#	Cell	Chromosome Pair 1 Allele	Chromosome Pair 2 Allele	Ploidy	# Chromosomes (each still duplicated)
1					
2					
3					

8. What separated during Anaphase I — homologous chromosomes or sister chromatids?

9. After Meiosis I, are the cells diploid or haploid? How many chromosomes does each cell have?

Part 4: Meiosis II — Separating Sister Chromatids

Perform the following steps for EACH of the two cells from Meiosis I.

Step 1: Metaphase II

1. In each cell, line up the duplicated chromosomes individually along the middle of the cell.

Step 2: Anaphase II

1. **Separate the sister chromatids** — pull them apart at the tape.
2. Move one chromatid to each side.

Step 3: Telophase II & Cytokinesis II

1. **Divide each cell** into two new cells.
2. You should now have **4 total cells** (gametes), each with **2 single chromosomes (n = 2)**.

Final Meiosis Results — All 4 Gametes

#	Cell	Gene A Allele	Gene B Allele	Genotype	Haploid?	# Chromosomes
1						
2						
3						
4						

10. What separated during Anaphase II — homologous chromosomes or sister chromatids?

11. How many gametes did you produce? Are they diploid or haploid?

12. Are any of the four gametes genetically identical to each other? Explain.

13. Why is it important that gametes are haploid (n) instead of diploid (2n)? What would happen at fertilization if both egg and sperm were diploid?

Part 5: Mitosis vs. Meiosis — Side-by-Side Comparison

Review your results and complete the comparison table.

Mitosis vs. Meiosis Comparison

#	Feature	Mitosis	Meiosis
1			
2			
3			
4			
5			
6			
7			

14. In mitosis, the parent cell (AaBb) produced daughter cells with genotype(s):

15. In meiosis, the parent cell (AaBb) produced gametes with genotype(s):

16. Which process produced more genetic variety? Why?

17. Summarize the key difference between mitosis and meiosis in your own words (2–3 sentences):

18. A human skin cell is diploid (2n = 46). If it undergoes mitosis, how many chromosomes will each daughter cell have? If a human reproductive cell (2n = 46) undergoes meiosis, how many chromosomes will each gamete have?

✂️ Chromosome Cutouts — Tear Out This Page

Cut along the dashed lines. **RED = Maternal** | **BLUE = Paternal**

Starting Cell Chromosomes

Use these 4 chromosomes to build your starting diploid cell (AaBb, 2n = 4).

MATERNAL	**PATERNAL**	**MATERNAL**	**PATERNAL**
Pair 1 · Gene A	Pair 1 · Gene A	Pair 2 · Gene B	Pair 2 · Gene B
A	**a**	**B**	**b**
(dominant)	(recessive)	(dominant)	(recessive)

S-Phase Duplicates (Sister Chromatids)

Keep these aside until S-Phase. Tape each duplicate to its matching chromosome above at the center to form sister chromatids.

MATERNAL	**PATERNAL**	**MATERNAL**	**PATERNAL**
Pair 1 · Copy	Pair 1 · Copy	Pair 2 · Copy	Pair 2 · Copy
A	**a**	**B**	**b**
(duplicate)	(duplicate)	(duplicate)	(duplicate)